



GRAPH

The GRAPH Courses

GENERATIVE AI FOR WORK & RESEARCH

NO-CODE AI TOOLS FOR RESPONSIBLE
TASK AUTOMATION

A 8-WEEK LIVE COHORT COURSE



Bootcamp Features



Live weekly workshops



Personalized feedback from expert instructors



Hands-on practice



Dedicated office hours for personalized support



ECTS credits for partner universities



Certificates of completion

Learning components

1

Pework

You will be assigned ~4 hours of prework tasks per week, to be completed before the workshop.



Lessons

Code along with the video tutorials and do practice exercises in RStudio



Online quizzes

Multiple choice theory quizzes and applied data quizzes

2

Workshops

Class time is used as a workshop where you engage in data analysis exercises that will help you apply the prework content and develop new skills while interacting with other students.



Whole-class teaching

Instructors demo new concepts



Group projects

Practice your skills with new datasets

Course Modules

OVERVIEW

MODULE 1

Documents,
Presentations
& Websites



MODULE 2

Automated
Data Analysis



MODULE 3

GenAI for
Research &
Text Analysis



MODULE 4

Custom GPTs
and Chatbots



MODULE 5

AI Tools for
Workflow
Automation



MODULE 6

GenAI for
Communication
& Outreach



MODULE 7

Capstone Project
& Showcase



MODULE 1

WEBSITES, DOCUMENTS & PRESENTATIONS

Learn how large language models (LLMs) work. Generate text responsibly for papers or grants, create project websites with no-code tools, format documents efficiently, and design clear presentations or posters using AI assistants.

Topics:

- How LLMs are trained and an overview of popular LLM providers
- Generating Markdown, LaTeX and HTML outputs
- Building simple research websites with ChatGPT Canvas and Replit
- Using Markdown for structured research documents
- Introduction to AI presentation tools for scientific communication

MODULE 2

AI-AUTOMATED DATA ANALYSIS

Master AI tools responsibly for analyzing data, manipulating files, and creating accurate, ethical automated reports. Use Julius, ChatGPT Code Interpreter and Windsurf to process datasets transparently, visualize results responsibly, and generate reports with integrity.

Topics:

- Use ChatGPT Code Interpreter for quick data analysis
- Build automated dashboards with TheBricks
- Process and transform datasets with Julius
- Implement batch file conversion and manipulation
- Create automated reports and data visualizations
- Design spreadsheet automations for recurring analyses

MODULE 3

GenAI for Research & Text Analysis

Learn AI techniques for literature search, text analysis, and knowledge synthesis. Explore how AI can responsibly accelerate information gathering, identify patterns in text data, and organize complex information. Develop systems for extracting insights from large text collections.

Topics:

- Conducting efficient literature reviews with Elicit and LiteRev
- Implementing text classification for document organization
- Applying sentiment analysis to identify trends and opinions
- Creating knowledge maps from document collections
- Designing automated summarization workflows
- Building information synthesis systems for research topics

MODULE 4

CUSTOM GPTS AND CHATBOTS

Build chatbots for research assistance, knowledge management, or specific task support. Explore RAG systems, API integrations, and Botpress workflows.

Topics:

- LLM chat architecture, context windows, and RAG systems
- Building custom GPTs for specific use cases
- Creating interactive chatbots with Botpress
- Integrating knowledge bases with your research materials

MODULE 5

AI FOR TASK AUTOMATION

Discover practical, responsible strategies for automating repetitive research tasks through AI-driven workflows. This module empowers you to design responsible AI agents for literature monitoring and orchestrate app-to-app workflows that save time.

Topics:

- Introduction to AI automation and agent-based systems
- Setting up email response automations
- Building AI agents for complex automation flows
- Browser control agents for web-based tasks

MODULE 6

GENAI FOR COMMUNICATION AND OUTREACH

Explore AI-driven approaches for research dissemination, focusing on creating authentic assets for sharing your findings responsibly. Learn to develop transparent methods for summarizing research, design accurate visuals for publications and presentations, and produce explanatory content with integrity and minimal manual effort.

Topics:

- AI-generated research summaries for different audiences
- Designing scientific figures and diagrams with AI tools
- AI-generated video abstracts and explanations
- Prompt engineering for scientific visualization
- Content scheduling for research communication

MODULE 7

CAPSTONE PROJECT & SHOWCASE

Collaborate with peers and instructors to create and refine a final AI solution, tool or presentation. Receive constructive feedback and address technical challenges thoughtfully. Conclude the program by presenting your final projects to an audience of peers and guests, demonstrating responsible AI practice.

Topics:

- Project progress check
- Peer feedback sessions
- Final project presentations
- Public Q&A session

Why GRAPH courses?

At GRAPH Courses, we're committed to making high-quality data science education accessible and impactful. Based at the University of Geneva, Switzerland, our program combines academic rigor with practical, industry-relevant skills.

Our bootcamp-style approach ensures an immersive learning experience. You'll gain technical skills and develop a portfolio of real-world projects, setting you apart in the job market.

What sets us apart is our supportive learning ecosystem. With live workshops, personalized code reviews, and a global community of peers, you're never alone. Our experienced instructors and mentors are dedicated to your success.

As a non-profit backed by academic institutions, we offer this comprehensive bootcamp at an affordable price without compromising on quality. Choose GRAPH Courses to invest in a transformative experience that will equip you with the skills and confidence to thrive in the data-driven world.

Our Partners



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Instructors & Mentors



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Senior Technical Business Analyst at Global Health Informatics Institute



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The GRAPH Courses



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